SpendWise Project Documentation

Team Details

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Introduction

- In today's digital world, managing personal finances has become increasingly important. Many people find it difficult to keep track of their daily expenses and savings. SpendWise is a backend application developed to help users manage and track their financial transactions effectively.
- The project is designed mainly for learning backend development using modern technologies such as Node.js, Express.js, MongoDB, and JWT authentication.
- It demonstrates how backend systems manage data, handle user authentication, and perform CRUD operations.

Abstract

- SpendWise is a backend-based expense tracking system developed to help users manage their financial transactions.
- The system allows users to register, log in securely using JWT authentication, and manage their expense records.

- Each user can add, update, view, and delete their transactions.

Objectives

- Develop a backend system for managing personal expenses.
- Implement secure user authentication using JWT.
- Perform CRUD operations on transaction data.
- Understand how REST APIs work.
- Learn backend development using Node.js and Express.js.
- Store and manage data using MongoDB.

Scope of the Project

- User registration and login system.
- Secure authentication using JWT tokens.
- Adding new transactions.
- Viewing transaction history.
- Updating existing transactions.
- Deleting transactions.

System Requirements

• Hardware: Computer or Laptop, Minimum 4GB RAM, Minimum 10GB free disk space, Internet connection. • Software: Windows / Linux / MacOS, Node.js, MongoDB, Visual Studio Code, Postman, Web Browser.

Tools and Technologies

- Node.js – runtime environment.
- Express.js – framework for building REST APIs.
- MongoDB – NoSQL database.

- JWT – secure authentication.
- Postman – API testing.
- Visual Studio Code – development environment.

System Architecture

- Client Layer – sends API requests.
- Application Layer – Node.js and Express handle logic and authentication.
- Database Layer – MongoDB stores user and transaction data.
- Flow: User → API Request → Express Server → MongoDB → Response.

Project Execution

- Setting up Node.js environment.
- Creating Express server.
- Connecting MongoDB database.
- Designing schemas for users and transactions.
- Implementing JWT authentication.
- Developing API routes.
- Testing APIs using Postman.

Limitations

- Backend only project.
- No graphical user interface.
- Limited visualization features.
- No advanced budgeting tools.

Future Enhancements

- Add frontend interface.
- Expense charts and analytics.
- Budget planning tools.
- Email notifications.
- Mobile application integration.
- Cloud deployment.

Conclusion

- SpendWise demonstrates backend development concepts including API creation, authentication, and database integration using Node.js, Express.js, MongoDB, and JWT.